

module4lab3

September 27, 2026

```
[1]: import pandas as pd
```

```
[2]: df=pd.read_csv("Downloads/truck.csv")
```

```
[3]: df.head()
```

```
[3]:      X      Y
0  12.4  11.2
1  14.3  12.5
2  14.5  12.7
3  14.9  13.1
4  16.1  14.1
```

```
[4]: df.shape
```

```
[4]: (23, 2)
```

```
[6]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 23 entries, 0 to 22
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype
---  ------  -
0    X      23 non-null      float64
1    Y      23 non-null      float64
dtypes: float64(2)
memory usage: 500.0 bytes
```

```
[7]: df.info
```

```
[7]: <bound method DataFrame.info of      X      Y
0  12.400000  11.200000
1  14.300000  12.500000
2  14.500000  12.700000
3  14.900000  13.100000
4  16.100000  14.100000
5  16.900000  14.800000
6  16.500000  14.400000
```

```

7    15.400000  13.400000
8    17.000000  14.900000
9    17.900000  15.600000
10   18.799999  16.400000
11   20.299999  17.700001
12   22.400000  19.600000
13   19.400000  16.900000
14   15.500000  14.000000
15   16.700001  14.600000
16   17.299999  15.100000
17   18.400000  16.100000
18   19.200001  16.799999
19   17.400000  15.200000
20   19.500000  17.000000
21   19.700001  17.200001
22   21.200001  18.600000>

```

```
[8]: df.describe()
```

```

[8]:
      count    23.000000    23.000000
      mean     17.465217    15.300000
      std       2.413901     2.057580
      min      12.400000    11.200000
      25%      15.800000    14.050000
      50%      17.299999    15.100000
      75%      19.300000    16.849999
      max      22.400000    19.600000

```

```
[9]: df.corr(numeric_only=True)
```

```

[9]:
      X      Y
X  1.00000  0.99854
Y  0.99854  1.00000

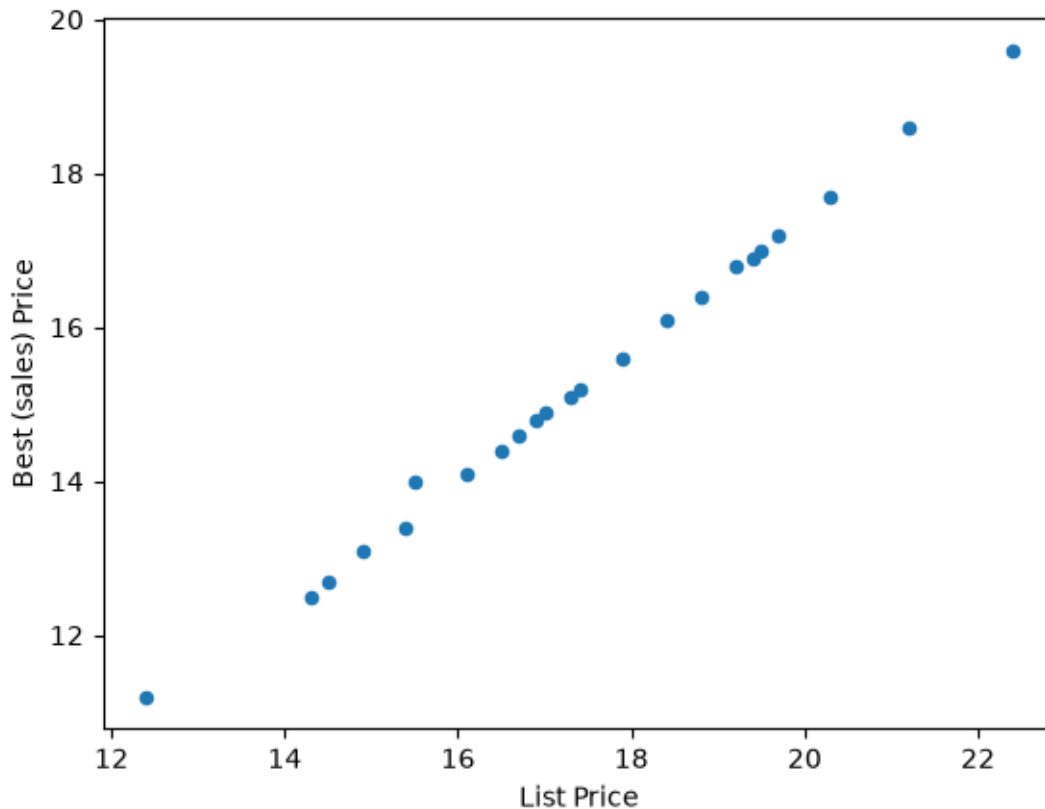
```

```

[20]: xlabel="List Price"
      ylabel="Best (sales) Price"
      df.plot.scatter(x="X", y="Y", xlabel=f"{xlabel}", ylabel=f"{ylabel}")

```

```
[20]: <Axes: xlabel='List Price', ylabel='Best (sales) Price'>
```



```
[ ]: #Markdown Cell
#the confidence of the truck and sale price is close to 1 meaning we can easily
  ↳predict the price of a truck and its sales price. ONLY issue is that we are
  ↳only sure of the values as long as the values fall within our data point.
  ↳Any values that might go out of the range of data might not fall into the
  ↳value making prediction difficult
```

```
[ ]: #Prompt Log
# "I want to label my axis like in excel is there a way for using a string to
  ↳help define the labels in the scatter plot?"
# I had to go back to module 1 reading and the tutorial from module 4 lab 1
  ↳tutorial 5 to use f{} and define to input labels for a clearer chart
```